

Certificate in Python Web Framework Development Assistant Assignment (CT290DS007)

By

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Version 01

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(A) Summary

(a) Purpose of this document:

習作要求：

- 為至少3個django資料模型建立不少於20筆樣本資料記錄。
- 產生一個獨立的python程式碼來：
 - a). 清理原始資料
 - b). 格式化資料集
 - c). 能將資料匯入和匯出到 django 資料庫
 - d). 最終結果可以在django管理面板下檢查

In English:

- (1) Create at least 20 sample data records for at least three Django models.
- (2) Develop Python code that can:
 - (2.1) Clean the raw data.
 - (2.2) Format the dataset.
 - (2.3) Import and export data to/from the Django database.
 - (2.4) Ensure the final result is visible and verifiable in the Django admin panel.

(b) Results: [All the requirements in \(A\) \(a\) above are fulfilled](#). For explanations, please refer to (B)(a) to (B)(e) below.

(c) Methodology:

- (1) Create a new [project](#) called [assignment](#) by cloning from ERB7 through “cp -r erb7 assignment” command.
- (2) Create a new app called [analytics](#) in the assignment project by “python manage.py startapp analytics” command.
- (3) Create a new database called assignment in Postgres by copying clinic database through “CREATE DATABASE assignment WITH TEMPLATE clinic OWNER postgres;” command.
- (4) Create local & remote repositories of GitHub & establish the link. The remote repository is called assignment.
- (5) In assignment/analytics/models.py, define three databases below:
 - (5.1) `class Financial(models.Model):`
 `year_month = models.CharField()`
 `revenue = models.IntegerField()`
 `expenses = models.IntegerField()`
 `profit_loss = models.IntegerField()`
 - (5.2) `class Operation(models.Model):`
 `year_month_date = models.CharField()`
 `manager_name = models.CharField()`
 `phone_number = models.CharField()`
 - (5.3) `class Asset(models.Model):`
 `description = models.CharField()`

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```

purchase_year_month = models.CharField()
cost = models.IntegerField()
accu_depre = models.IntegerField()
net_book_value = models.IntegerField()

```

Remarks:

- The Financial class is to record monthly profit and loss data of a company.
- The Operation class is to record names & phone numbers of responsible managers of a company.
- The Asset class is a fixed assets register for a company.

(6) Register the app analytics and amend parameters in assignment/config/settings.py as:

```

INSTALLED_APPS = [
...
'analytics.apps.AnalyticsConfig',
...
]

```

```

DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.postgresql',
        'NAME': 'assignment',
        'USER': 'postgres',
        'PASSWORD': '12345',
        'HOST': 'localhost',
        'PORT': '5432'
    }
}

```

(7) Create and apply migrations by

- python manage.py makemigrations
- python manage.py migrate

(8) Enable admin interface. In assignment/analytics/admin.py, execute the following code
(This makes all three databases manageable via Django Admin):

```

from django.contrib import admin
from .models import Financial, Operation, Asset

class FinancialAdmin(admin.ModelAdmin):
    list_display = ('year_month', 'revenue_thousand', 'expenditure_thousand',
                    'profit_loss_thousand')
    list_per_page = 25

class OperationAdmin(admin.ModelAdmin):
    list_display = ('year_month_day', 'manager_name', 'phone_number')
    list_per_page = 25

```

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```
class AssetAdmin(admin.ModelAdmin):
    list_display = ('description', 'purchase_year_month', 'cost_million',
                   'accu_depre_million', 'net_book_value_million')
    list_per_page = 25

admin.site.register(Financial, FinancialAdmin)
admin.site.register(Operation, OperationAdmin)
admin.site.register(Asset, AssetAdmin)
```

- (9) Add "export data" function to built-in admin interface by including the code below in assignment/analytics/admin.py:

```
import csv
from django.http import HttpResponse
def export_as_csv(modeladmin, request, queryset):
    meta = modeladmin.model._meta
    field_names = [field.name for field in meta.fields]
    response = HttpResponse(content_type='text/csv')
    response['Content-Disposition'] = f'attachment; filename={meta}.csv'
    writer = csv.writer(response)
    writer.writerow(field_names)
    for obj in queryset:
        writer.writerow([getattr(obj, field) for field in field_names])
    return response
export_as_csv.short_description = "Export Selected to CSV"
actions = [export_as_csv]
```

- (10) Add "import data" function to the built-in admin interface by:
- (10.1) Creating a Simple CSV Upload Form. For source code, please refer to Appendix (C)(b) assignment/analytics/forms.py.
 - (10.2) Adding Import Logic to Admin Classes. For source code, please refer to Appendix (C)(a) assignment/analytics/admin.py.
 - (10.3) Extending Admin Classes with Import View. For source code, please refer to Appendix (C)(a) assignment/analytics/admin.py.
 - (10.4) Creating the Upload Form Template. For source code, please refer to Appendix (C)(e) assignment/templates/admin/csv_form.html.
 - (10.5) Adding Import Button to Admin Interface. For source code, please refer to Appendix (C) (f), (g), (h) i.e.:
 - assignment/templates/admin/analytics/financial/change_list.html
 - assignment/templates/admin/analytics/operation/change_list.html
 - assignment/templates/admin/analytics/asset/change_list.html
- (11) Ensure datasets are formatted.
- (12) Ensure cleaning raw data function is available.

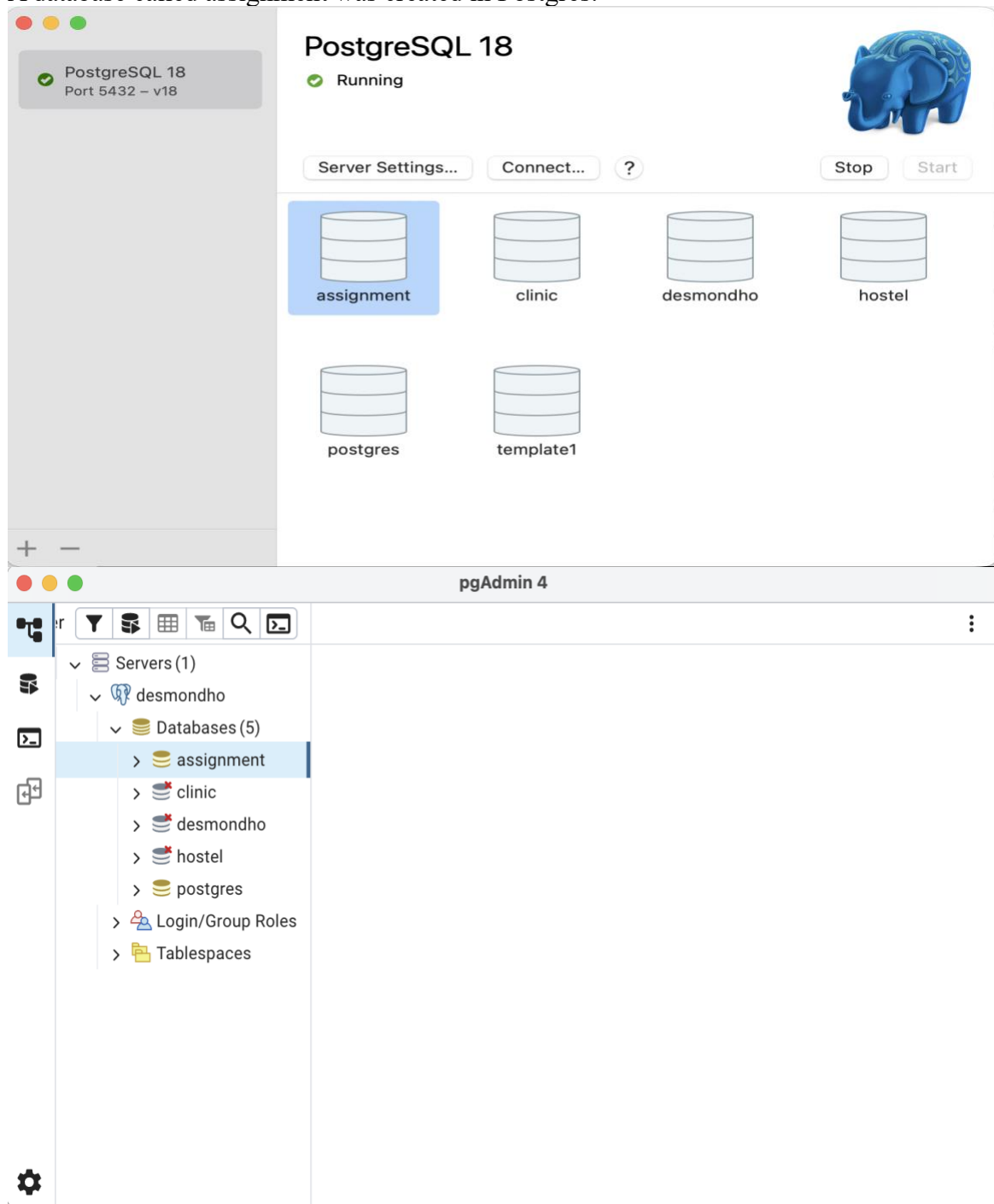
- (d) Source Files: For all the source files please refer to the attached zipped file called [assignment.zip](#) or visit the following site:
<https://github.com/hosiuhangdesmond/assignment>

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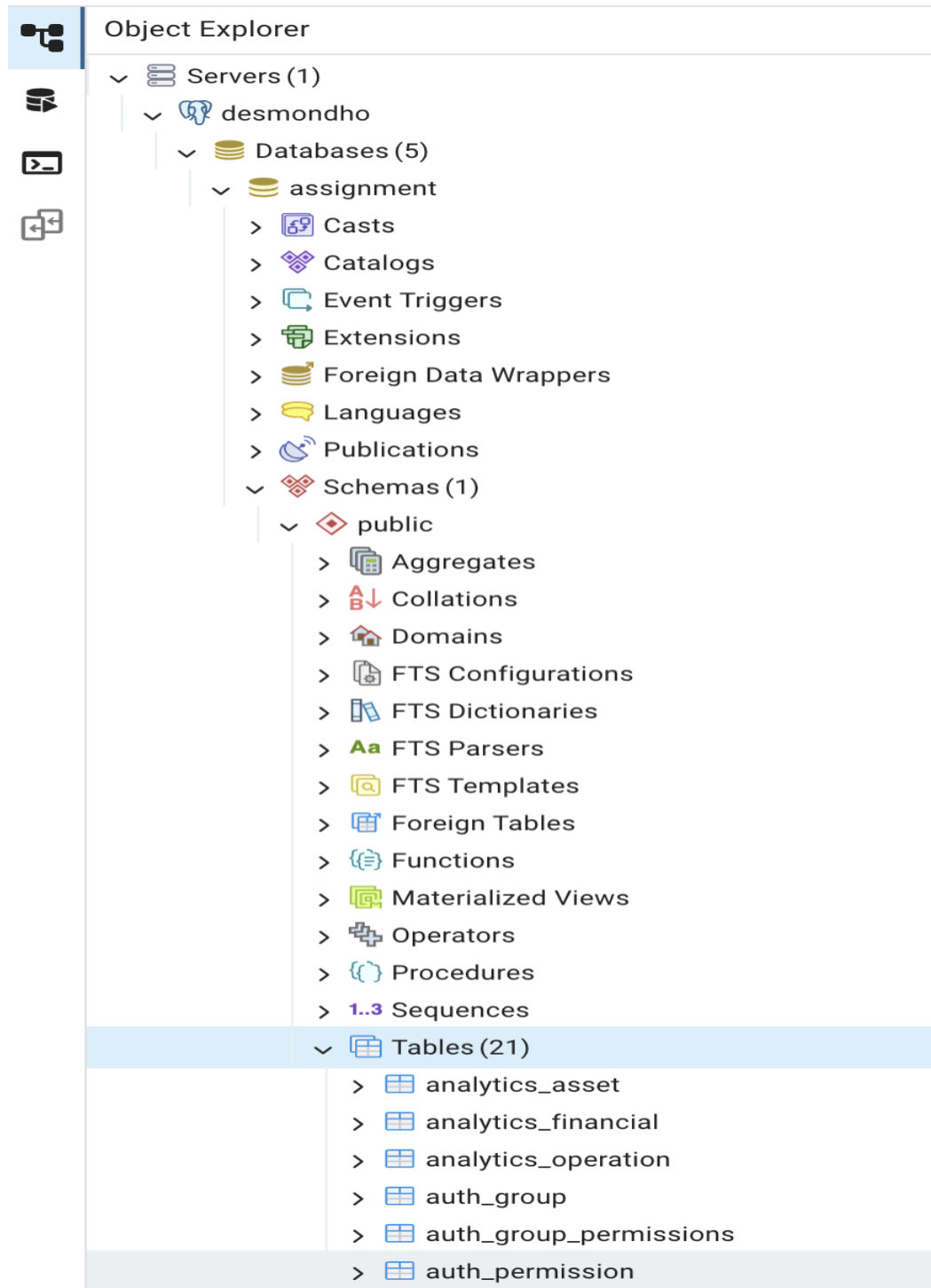
(B) Explanations

(a) Create at least 20 sample data records for at least three Django models

(1) A database called assignment was created in Postgres:



(2) Three tables were created under the assignment database: analytics_asset, analytics_financial and analytics_operation:



(3) Each table has more than 20 sample data records.

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The screenshot displays a database management interface. On the left, the 'Object Explorer' shows a tree of database objects, with 'analytics_asset' selected under the 'Tables (21)' category. The main pane shows the 'Query' editor with the following SQL query:

```
1 SELECT * FROM public.analytics_asset
2 ORDER BY id ASC
```

Below the query editor, the 'Data Output' tab shows the results of the query. The table has 22 rows, showing rows 1 to 22. The columns are: id [PK] bigint, description character varying, purchase_year_month character varying, cost_million integer, accu_depre_million integer, and net_book_value_million integer.

	id [PK] bigint	description character varying	purchase_year_month character varying	cost_million integer	accu_depre_million integer	net_book_value_million integer
1	1	BuildingA	2005_12	5	2	3
2	2	BuildingB	2015_07	10	3	7
3	3	BuildingC	2010_09	5	3	2
4	4	ComputerSysA	2017_03	7	5	2
5	5	ComputerSysB	2007_09	3	3	0
6	6	CarA	1994_06	4	3	1
7	7	CarB	2021_01	3	1	2
8	8	CarC	2000_04	7	3	4
9	9	CarD	2001_03	2	2	0
10	10	CarE	2020_08	6	4	2
11	11	ComputerSysC	2014_09	20	13	7
12	12	ComputerSysD	1986_05	40	40	0
13	13	ComputerSysE	2022_01	30	13	17
14	14	BuildingD	2000_08	60	40	20
15	15	BuildingE	2013_09	200	130	70
16	16	MachineA	2001_03	2	2	0
17	17	MachineB	2020_08	60	40	20
18	18	MachineC	2014_09	20	13	7
19	19	MachineD	1986_05	20	20	0
20	20	MachineE	2022_01	300	130	170
21	21	FurnitureA	1967_04	2	2	0
22	22	FurnitureB	1977_05	3	3	0

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The screenshot displays a database management interface. On the left, the 'Object Explorer' shows a tree of database objects, with 'analytics_financial' selected under the 'Tables (21)' category. The main pane shows the 'Query' editor with the following SQL query:

```
SELECT * FROM public.analytics_financial
ORDER BY id ASC
```

Below the query editor, the 'Data Output' tab shows the results of the query. The table has 22 rows, showing data from 2024 to 2025. The columns are: id [PK] bigint, year_month character varying, revenue_thousand integer, expenditure_thousand integer, and profit_loss_thousand integer. The data is as follows:

id [PK] bigint	year_month character varying	revenue_thousand integer	expenditure_thousand integer	profit_loss_thousand integer
1	2024_01	25000	19000	6000
2	2024_02	50000	30000	20000
3	2024_03	15000	20000	-5000
4	2024_04	80000	70000	10000
5	2024_05	90000	65000	25000
6	2024_06	50000	60000	-10000
7	2024_07	45000	50000	-5000
8	2024_08	85000	75000	10000
9	2024_09	45000	65000	-20000
10	2024_10	75000	50000	25000
11	2024_11	100000	80000	20000
12	2024_12	150000	110000	40000
13	2025_01	160000	120000	40000
14	2025_02	200000	150000	50000
15	2025_03	180000	140000	40000
16	2025_04	170000	140000	30000
17	2025_05	160000	130000	30000
18	2025_06	150000	130000	20000
19	2025_07	140000	120000	20000
20	2025_08	130000	120000	10000
21	2025_09	120000	110000	10000
22	2025_10	110000	100000	10000

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The screenshot displays a database management interface. On the left, the 'Object Explorer' shows a tree of database objects, with 'analytics_operation' selected under the 'Tables (21)' category. The main pane shows the 'Query' tab with the following SQL query:

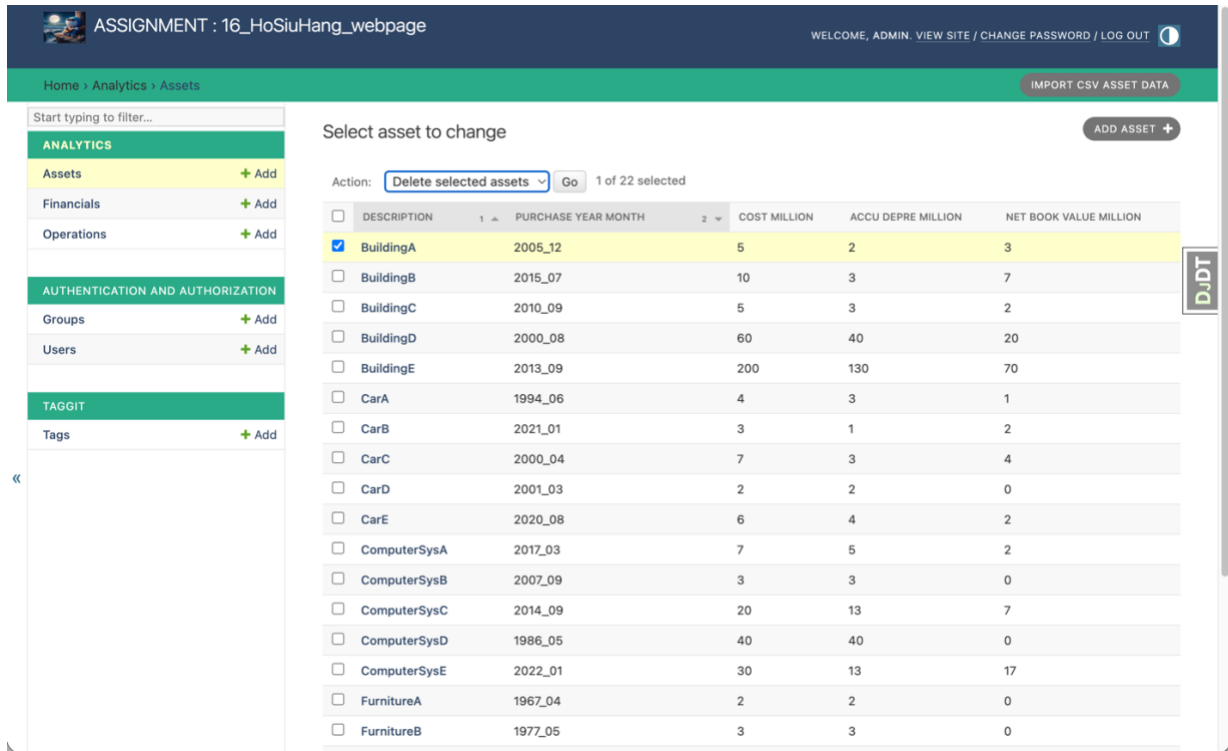
```
SELECT * FROM public.analytics_operation
ORDER BY id ASC
```

Below the query, the 'Data Output' tab shows the results of the query, displaying 22 rows. The columns are: id [PK] bigint, year_month_day character varying, manager_name character varying, and phone_number character varying. The data is sorted by id in ascending order.

	id [PK] bigint	year_month_day character varying	manager_name character varying	phone_number character varying
1	1	2025_07_01	ChanTaiMan	9876-5432
2	2	2025_07_02	WongTaiSin	7654-2345
3	3	2025_07_03	YiuTongHui	3645-7896
4	4	2025_07_04	TamLiYi	3401-6633
5	5	2025_07_05	YuHoSam	5693-5239
6	6	2025_07_06	HuangJi	8423-0759
7	7	2025_07_07	ZhouXinXin	7734-9922
8	8	2025_07_08	YuHoSam	5693-5239
9	9	2025_07_09	TamLiYi	3401-6633
10	10	2025_07_10	ZhouXinXin	7734-9922
11	11	2025_07_11	YiuTongHui	3645-7896
12	12	2025_07_12	ChanTaiMan	9876-5432
13	13	2025_07_13	TamLiYi	3401-6633
14	14	2025_07_14	YuHoSam	5693-5239
15	15	2025_07_15	HuangJi	8423-0759
16	16	2025_07_16	ChanTaiMan	9876-5432
17	17	2025_07_17	WongTaiSin	7654-2345
18	18	2025_07_18	HuangJi	8423-0759
19	19	2025_07_19	ZhouXinXin	7734-9922
20	20	2025_07_20	ChanTaiMan	9876-5432
21	21	2025_07_21	ZhouXinXin	7734-9922
22	22	2025_07_22	YuHoSam	5693-5239

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- (b) Develop Python code that can clean the raw data
- (1) By using Django, there is a built-in delete option: "delete_selected.short_description = _("Delete selected %(verbose_name_plural)s)". It is part of the internal source code of Django and is located at: `django/contrib/admin/actions.py`
- (2) Select the record to be cleaned/deleted e.g. BuildingA. Choose “Delete selected assets” action. Press “Go”.



ASSIGNMENT : 16_HoSiuhang_webpage

WELCOME, ADMIN. [VIEW SITE](#) / [CHANGE PASSWORD](#) / [LOG OUT](#)

Home > Analytics > Assets

Start typing to filter...

ANALYTICS

Assets [+ Add](#)

Financials [+ Add](#)

Operations [+ Add](#)

AUTHENTICATION AND AUTHORIZATION

Groups [+ Add](#)

Users [+ Add](#)

TAGGIT

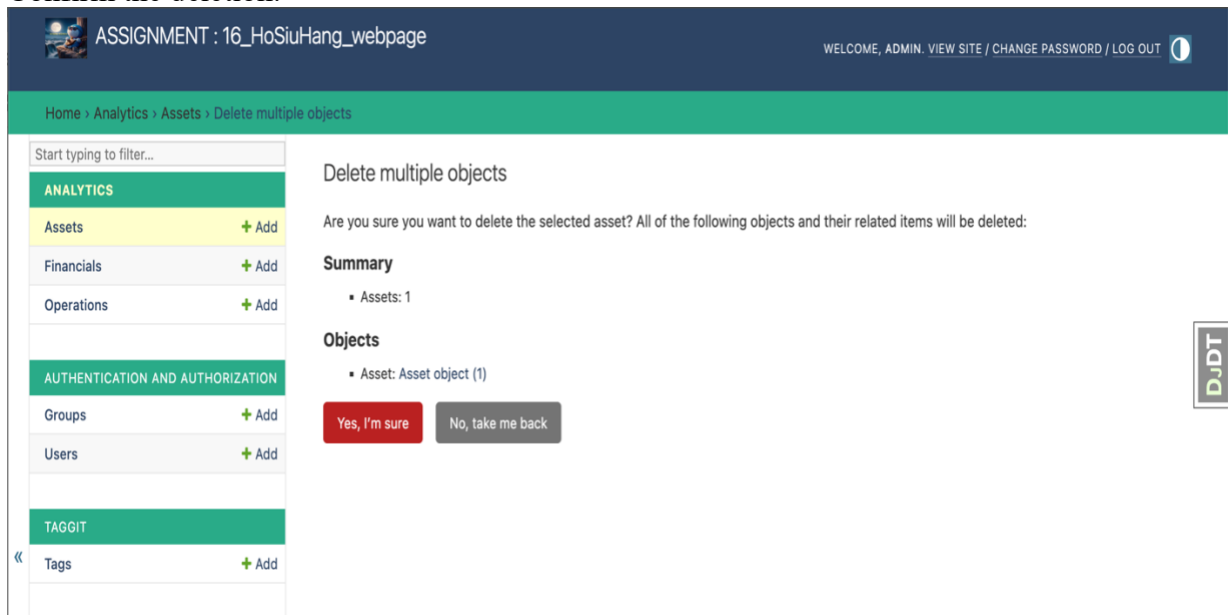
Tags [+ Add](#)

Select asset to change

Action: **Delete selected assets** Go 1 of 22 selected

☐	DESCRIPTION	PURCHASE YEAR MONTH	COST MILLION	ACCU DEPRE MILLION	NET BOOK VALUE MILLION
☒	BuildingA	2005_12	5	2	3
☐	BuildingB	2015_07	10	3	7
☐	BuildingC	2010_09	5	3	2
☐	BuildingD	2000_08	60	40	20
☐	BuildingE	2013_09	200	130	70
☐	CarA	1994_06	4	3	1
☐	CarB	2021_01	3	1	2
☐	CarC	2000_04	7	3	4
☐	CarD	2001_03	2	2	0
☐	CarE	2020_08	6	4	2
☐	ComputerSysA	2017_03	7	5	2
☐	ComputerSysB	2007_09	3	3	0
☐	ComputerSysC	2014_09	20	13	7
☐	ComputerSysD	1986_05	40	40	0
☐	ComputerSysE	2022_01	30	13	17
☐	FurnitureA	1967_04	2	2	0
☐	FurnitureB	1977_05	3	3	0

- (3) Confirm the deletion.



ASSIGNMENT : 16_HoSiuhang_webpage

WELCOME, ADMIN. [VIEW SITE](#) / [CHANGE PASSWORD](#) / [LOG OUT](#)

Home > Analytics > Assets > Delete multiple objects

Start typing to filter...

ANALYTICS

Assets [+ Add](#)

Financials [+ Add](#)

Operations [+ Add](#)

AUTHENTICATION AND AUTHORIZATION

Groups [+ Add](#)

Users [+ Add](#)

TAGGIT

Tags [+ Add](#)

Delete multiple objects

Are you sure you want to delete the selected asset? All of the following objects and their related items will be deleted:

Summary

- Assets: 1

Objects

- Asset: Asset object (1)

[Yes, I'm sure](#) [No, take me back](#)

- (4) Then the record (BuildingA) is deleted. The other two tables have the same deletion option.

Start typing to filter...

ANALYTICS

Assets + Add

Financials + Add

Operations + Add

AUTHENTICATION AND AUTHORIZATION

Groups + Add

Users + Add

TAGGIT

Tags + Add

Successfully deleted 1 asset.

IMPORT CSV ASSET DATA

ADD ASSET +

Select asset to change

Action: ----- Go 0 of 21 selected

☐	DESCRIPTION	1 PURCHASE YEAR MONTH 2	COST MILLION	ACCU DEPRE MILLION	NET BOOK VALUE MILLION
☐	BuildingB	2015_07	10	3	7
☐	BuildingC	2010_09	5	3	2
☐	BuildingD	2000_08	60	40	20
☐	BuildingE	2013_09	200	130	70
☐	CarA	1994_06	4	3	1
☐	CarB	2021_01	3	1	2
☐	CarC	2000_04	7	3	4
☐	CarD	2001_03	2	2	0
☐	CarE	2020_08	6	4	2

DjDT

- (c) Develop Python code that can format the dataset
All tables are stored in Postgres database in formatted style. All data exported from Postgres database are in CSV format. If external data are imported into the Postgres, they must be in formatted style i.e. CSV format. Please refer to the source code in (C) Appendixes.
- (d) Develop Python code that can import and export data to/from the Django database
Please refer to the source code in (C) Appendixes.
- (1) Import:

For operation table, create five dummy records in file operation_20251101.txt:

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```

operation_20251101.txt
year_month_day,manager_name,phone_number
2025_07_23,HuangJi,8423-0759
2025_07_24,ZhouXinXin,7734-9922
2025_07_25,YuHoSam,5693-5239
2025_07_26,TamLiYi,3401-6633
2025_07_27,YiuTongHui,3645-7896

```

Before import

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Home > Analytics > Operations IMPORT CSV OPERATION DATA

Start typing to filter...

ANALYTICS

Assets + Add

Financials + Add

Operations + Add

AUTHENTICATION AND AUTHORIZATION

Groups + Add

Users + Add

TAGGIT

Tags + Add

Select operation to change ADD OPERATION +

Action: ----- Go 0 of 22 selected

☐	YEAR MONTH DAY	MANAGER NAME	PHONE NUMBER
☐	2025_07_22	YuHoSam	5693-5239
☐	2025_07_21	ZhouXinXin	7734-9922
☐	2025_07_20	ChanTaiMan	9876-5432
☐	2025_07_19	ZhouXinXin	7734-9922
☐	2025_07_18	HuangJi	8423-0759
☐	2025_07_17	WongTaiSin	7654-2345
☐	2025_07_16	ChanTaiMan	9876-5432
☐	2025_07_15	HuangJi	8423-0759
☐	2025_07_14	YuHoSam	5693-5239
☐	2025_07_13	TamLiYi	3401-6633
☐	2025_07_12	ChanTaiMan	9876-5432

Press “IMPORT CSV OPERATION DATA” button. Choose file. Press “Import” button.

ASSIGNMENT : 16_HoSiuHang_webpage

Home > Import CSV

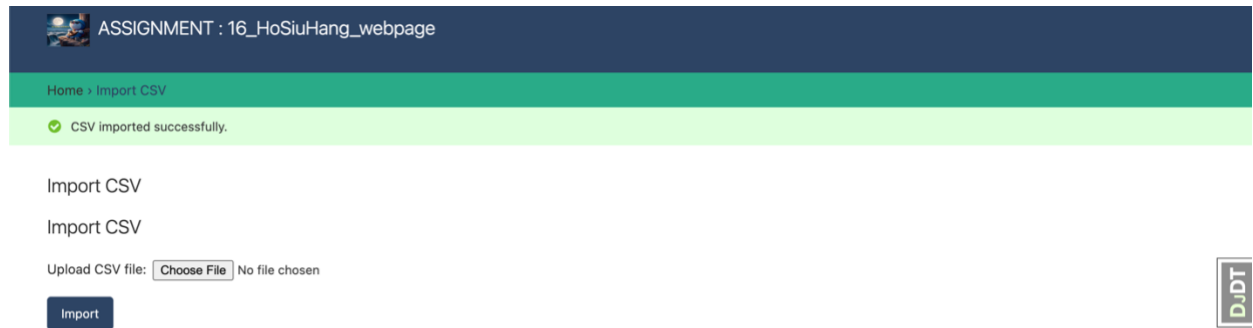
Import CSV

Import CSV

Upload CSV file: Choose File operation_20251101.txt

Import

The five records are successfully imported into the operation table.

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ASSIGNMENT : 16_HoSiuHang_webpage

Home > Import CSV

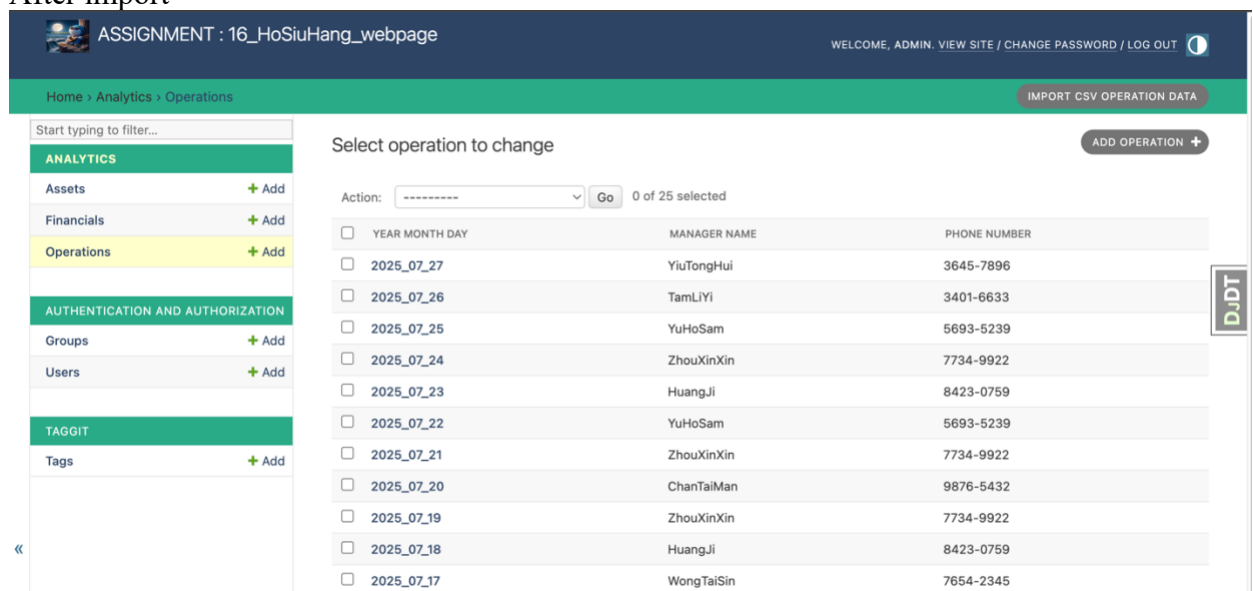
✓ CSV imported successfully.

Import CSV

Import CSV

Upload CSV file: No file chosen

After import



ASSIGNMENT : 16_HoSiuHang_webpage

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Home > Analytics > Operations

IMPORT CSV OPERATION DATA

Start typing to filter...

ANALYTICS

Assets [+ Add](#)

Financials [+ Add](#)

Operations [+ Add](#)

AUTHENTICATION AND AUTHORIZATION

Groups [+ Add](#)

Users [+ Add](#)

TAGGIT

Tags [+ Add](#)

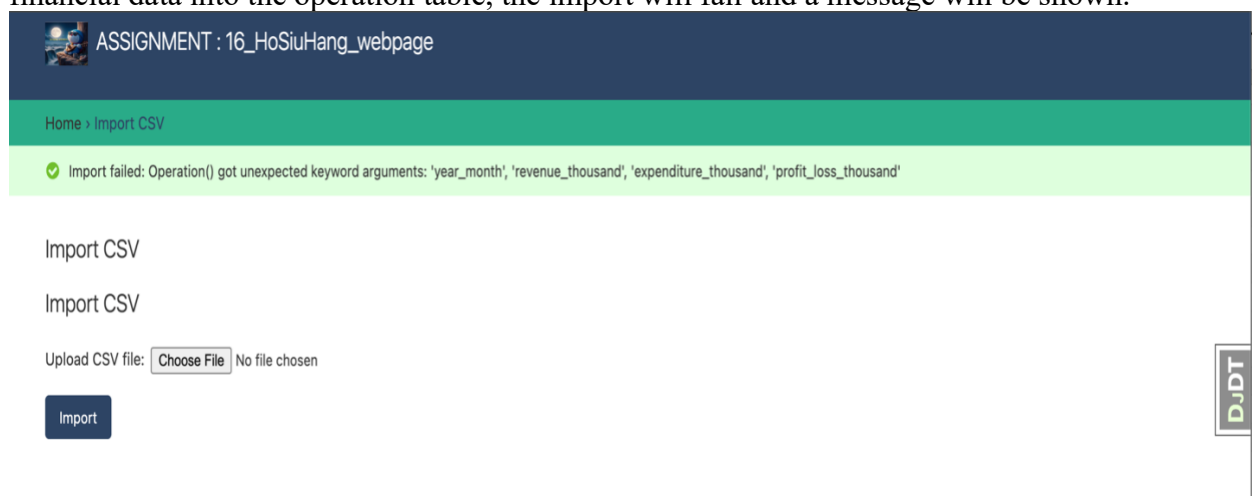
Select operation to change

Action: Go 0 of 25 selected

☐	YEAR MONTH DAY	MANAGER NAME	PHONE NUMBER
☐	2025_07_27	YiuTongHui	3645-7896
☐	2025_07_26	TamLIYi	3401-6633
☐	2025_07_25	YuHoSam	5693-5239
☐	2025_07_24	ZhouXinXin	7734-9922
☐	2025_07_23	HuangJi	8423-0759
☐	2025_07_22	YuHoSam	5693-5239
☐	2025_07_21	ZhouXinXin	7734-9922
☐	2025_07_20	ChanTaiMan	9876-5432
☐	2025_07_19	ZhouXinXin	7734-9922
☐	2025_07_18	HuangJi	8423-0759
☐	2025_07_17	WongTaiSin	7654-2345

ADD OPERATION +

If the format of the source file does not match with that of the destination file e.g. import financial data into the operation table, the import will fail and a message will be shown.



ASSIGNMENT : 16_HoSiuHang_webpage

Home > Import CSV

✓ Import failed: Operation() got unexpected keyword arguments: 'year_month', 'revenue_thousand', 'expenditure_thousand', 'profit_loss_thousand'

Import CSV

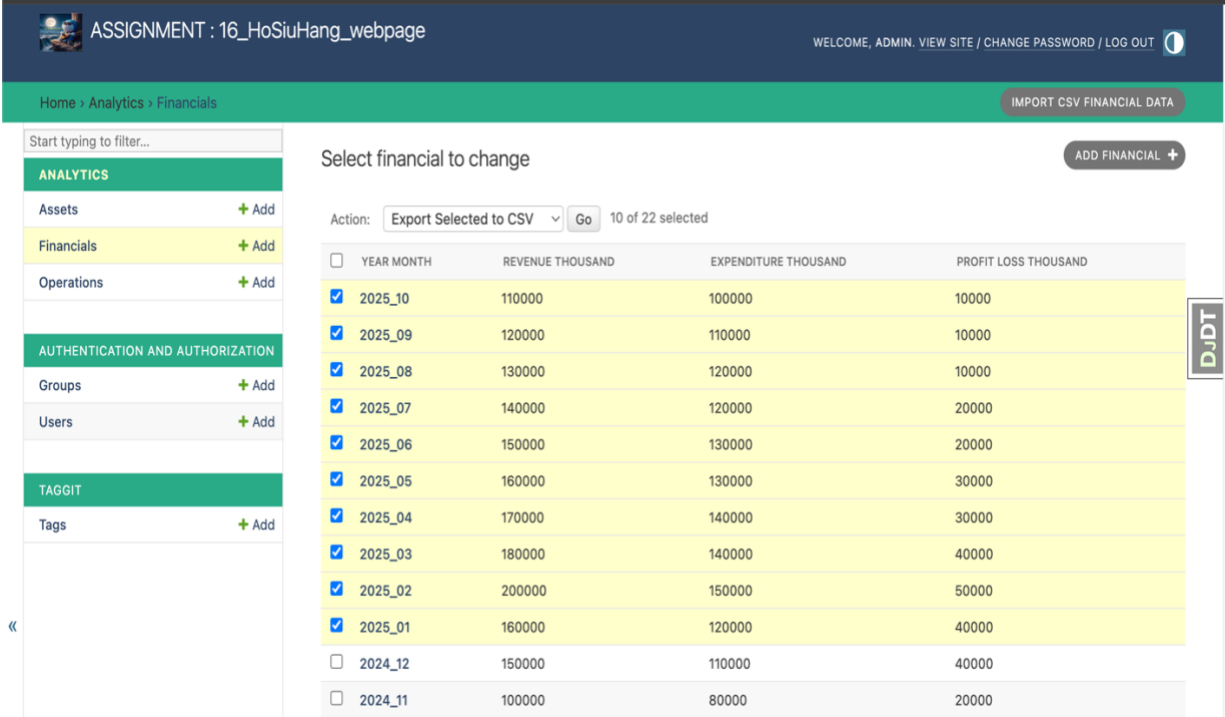
Import CSV

Upload CSV file: No file chosen

The import function is also available in the other two tables.

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Select the records to export e.g. all 2025 records. Choose “Export Selected to CSV” action. Press “Go”.



ASSIGNMENT : 16_HoSiuHang_webpage

WELCOME, ADMIN. [VIEW SITE](#) / [CHANGE PASSWORD](#) / [LOG OUT](#)

Home > Analytics > Financials

IMPORT CSV FINANCIAL DATA

Start typing to filter...

ANALYTICS

- Assets + Add
- Financials + Add
- Operations + Add

AUTHENTICATION AND AUTHORIZATION

- Groups + Add
- Users + Add

TAGGIT

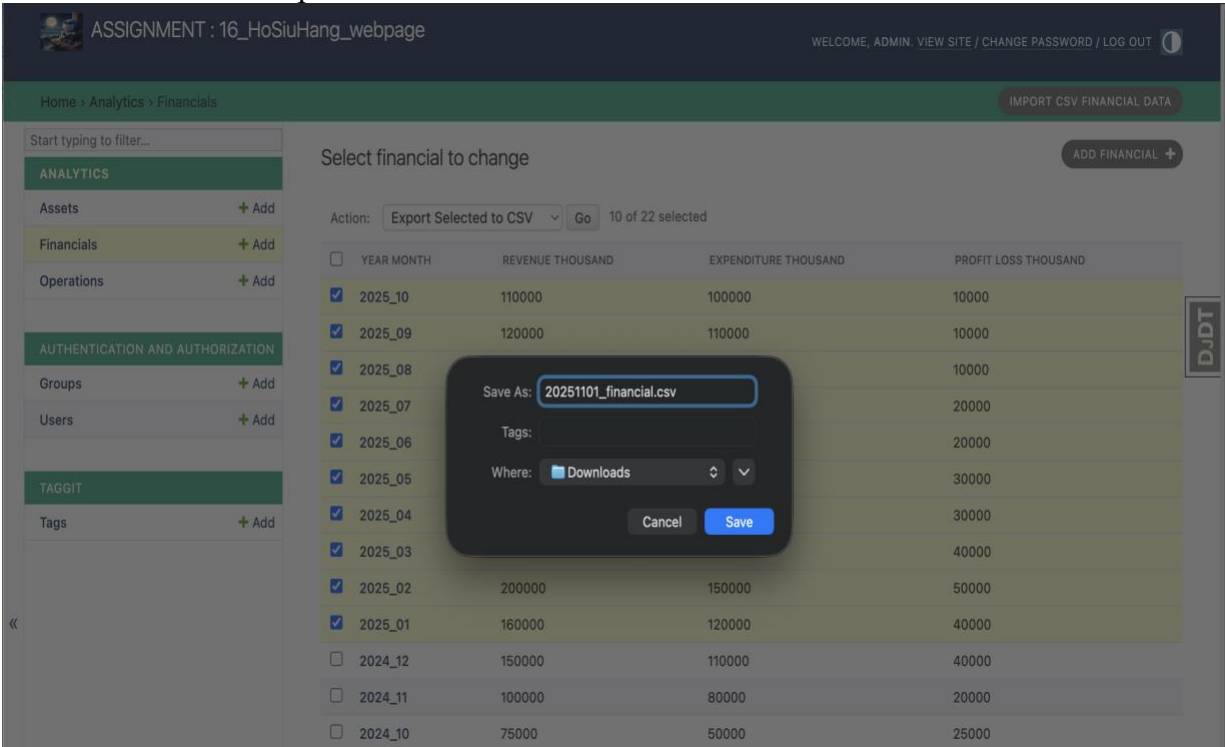
- Tags + Add

Select financial to change

Action: **Export Selected to CSV** Go 10 of 22 selected

☐	YEAR MONTH	REVENUE THOUSAND	EXPENDITURE THOUSAND	PROFIT LOSS THOUSAND
☒	2025_10	110000	100000	10000
☒	2025_09	120000	110000	10000
☒	2025_08	130000	120000	10000
☒	2025_07	140000	120000	20000
☒	2025_06	150000	130000	20000
☒	2025_05	160000	130000	30000
☒	2025_04	170000	140000	30000
☒	2025_03	180000	140000	40000
☒	2025_02	200000	150000	50000
☒	2025_01	160000	120000	40000
☐	2024_12	150000	110000	40000
☐	2024_11	100000	80000	20000

Provide name of the exported file and destination. Press “Save” button.



ASSIGNMENT : 16_HoSiuHang_webpage

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Home > Analytics > Financials

IMPORT CSV FINANCIAL DATA

Start typing to filter...

ANALYTICS

- Assets + Add
- Financials + Add
- Operations + Add

AUTHENTICATION AND AUTHORIZATION

- Groups + Add
- Users + Add

TAGGIT

- Tags + Add

Select financial to change

Action: **Export Selected to CSV** Go 10 of 22 selected

☐	YEAR MONTH	REVENUE THOUSAND	EXPENDITURE THOUSAND	PROFIT LOSS THOUSAND
☒	2025_10	110000	100000	10000
☒	2025_09	120000	110000	10000
☒	2025_08	130000	120000	10000
☒	2025_07	140000	120000	20000
☒	2025_06	150000	130000	20000
☒	2025_05	160000	130000	30000
☒	2025_04	170000	140000	30000
☒	2025_03	180000	140000	40000
☒	2025_02	200000	150000	50000
☒	2025_01	160000	120000	40000
☐	2024_12	150000	110000	40000
☐	2024_11	100000	80000	20000
☐	2024_10	75000	50000	25000

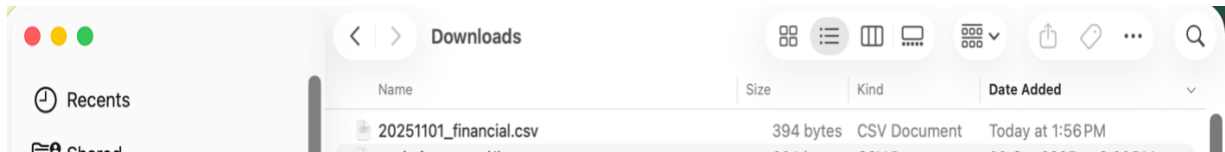
Save As: **20251101_financial.csv**

Tags:

Where: **Downloads**

Cancel Save

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The selected records are successfully exported in a file called 20251101_financial.csv. This export function is also available in the other two tables.

id	year_month	revenue_thousand	expenditure_thousand	profit_loss_thousand
22	2025_10	110000	100000	10000
21	2025_09	120000	110000	10000
20	2025_08	130000	120000	10000
19	2025_07	140000	120000	20000
18	2025_06	150000	130000	20000
17	2025_05	160000	130000	30000
16	2025_04	170000	140000	30000
15	2025_03	180000	140000	40000
14	2025_02	200000	150000	50000
13	2025_01	160000	120000	40000

- (e) Develop Python code that can ensure the final result is visible and verifiable in the Django admin panel

Done. Please refer to the source code in (C) Appendixes.

(C) Appendixes: source code

(a) assignment/analytics/admin.py

```
import csv
from django.http import HttpResponse
from django.contrib import admin, messages
from django.urls import path
from django.shortcuts import render, redirect
from .models import Financial, Operation, Asset
from .forms import SimpleCSVImportForm

def simple_csv_import(modeladmin, request, model):
    if request.method == "POST":
        form = SimpleCSVImportForm(request.POST, request.FILES)
        if form.is_valid():
            csv_file = form.cleaned_data['csv_file']
            decoded_file = csv_file.read().decode('utf-8').splitlines()
            reader = csv.DictReader(decoded_file)

            try:
                for row in reader:
                    model.objects.create(**row)
                messages.success(request, "CSV imported successfully.")
            except Exception as e:
                messages.error(request, f'Import failed: {str(e)}')
            return redirect(request.path)
        else:
            form = SimpleCSVImportForm()

    context = {
        'form': form,
        'title': 'Import CSV',
        'opts': model._meta,
    }
    return render(request, "admin/csv_form.html", context)

def export_as_csv(modeladmin, request, queryset):
    meta = modeladmin.model._meta
    field_names = [field.name for field in meta.fields]

    response = HttpResponse(content_type='text/csv')
    response['Content-Disposition'] = f'attachment; filename={meta}.csv'
    writer = csv.writer(response)

    writer.writerow(field_names)
    for obj in queryset:
        writer.writerow([getattr(obj, field) for field in field_names])

    return response
```

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```
export_as_csv.short_description = "Export Selected to CSV"

class FinancialAdmin(admin.ModelAdmin):
    list_display = ('year_month', 'revenue_thousand', 'expenditure_thousand',
'profit_loss_thousand')
    list_per_page = 25
    actions = [export_as_csv]

    change_list_template = "admin/analytics/financial/change_list.html"

    def get_urls(self):
        urls = super().get_urls()
        custom_urls = [
            path('import-csv/', self.admin_site.admin_view(self.import_csv),
name='financial_import_csv'),
        ]
        return custom_urls + urls

    def import_csv(self, request):
        return simple_csv_import(self, request, Financial)

class OperationAdmin(admin.ModelAdmin):
    list_display = ('year_month_day', 'manager_name', 'phone_number')
    list_per_page = 25
    actions = [export_as_csv]

    change_list_template = "admin/analytics/operation/change_list.html"

    def get_urls(self):
        urls = super().get_urls()
        custom_urls = [
            path('import-csv/', self.admin_site.admin_view(self.import_csv),
name='operation_import_csv'),
        ]
        return custom_urls + urls

    def import_csv(self, request):
        return simple_csv_import(self, request, Operation)

class AssetAdmin(admin.ModelAdmin):
    list_display = ('description', 'purchase_year_month', 'cost_million', 'accu_depre_million',
'net_book_value_million')
    list_per_page = 25
    actions = [export_as_csv]

    change_list_template = "admin/analytics/asset/change_list.html"

    def get_urls(self):
        urls = super().get_urls()
```

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```

        custom_urls = [
            path('import-csv/', self.admin_site.admin_view(self.import_csv),
name='asset_import_csv'),
        ]
        return custom_urls + urls

def import_csv(self, request):
    return simple_csv_import(self, request, Asset)

admin.site.register(Financial, FinancialAdmin)
admin.site.register(Operation, OperationAdmin)
admin.site.register(Asset, AssetAdmin)

```

- (b) [assignment/analytics/forms.py](#)
from django import forms

```

class SimpleCSVImportForm(forms.Form):
    csv_file = forms.FileField(label="Upload CSV file")

```

- (c) [assignment/analytics/models.py](#)
from django.db import models

```

class Financial(models.Model):
    year_month = models.CharField()
    revenue_thousand = models.IntegerField()
    expenditure_thousand = models.IntegerField()
    profit_loss_thousand = models.IntegerField()

class Operation(models.Model):
    year_month_day = models.CharField()
    manager_name = models.CharField()
    phone_number = models.CharField()

class Asset(models.Model):
    description = models.CharField()
    purchase_year_month = models.CharField()
    cost_million = models.IntegerField()
    accu_depre_million = models.IntegerField()
    net_book_value_million = models.IntegerField()

```

- (d) [assignment/config/settings.py/databases](#)

```

DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.postgresql',
        'NAME': 'assignment',
        'USER': 'postgres',

```

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```

        'PASSWORD': '12345',
        'HOST': 'localhost',
        'PORT': '5432'
    }
}

```

- (e) [assignment/templates/admin/csv_form.html](#)
- ```

{% extends "admin/base_site.html" %}
{% block content %}
<h1>{{ title }}</h1>
<form method="post" enctype="multipart/form-data">
 {% csrf_token %}
 {{ form.as_p }}
 <input type="submit" value="Import">
</form>
{% endblock %}

```
- (f) [assignment/templates/admin/analytics/financial/change\\_list.html](#)
- ```

{% extends "admin/change_list.html" %}
{% block object-tools %}
<div class="object-tools">
    <ul class="object-tools">
        <li><a href="{% url 'admin:financial_import_csv' %}">Import CSV financial
data</a></li>
    </ul>
</div>
{{ block.super }}
{% endblock %}

```
- (g) [assignment/templates/admin/analytics/operation/change_list.html](#)
- ```

{% extends "admin/change_list.html" %}
{% block object-tools %}
<div class="object-tools">
 <ul class="object-tools">
 Import CSV operation
data

</div>
{{ block.super }}
{% endblock %}

```
- (h) [assignment/templates/admin/analytics/asset/change\\_list.html](#)
- ```

{% extends "admin/change_list.html" %}
{% block object-tools %}
<div class="object-tools">

```

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```
<ul class="object-tools">
  <li><a href="{% url 'admin:asset_import_csv' %}">Import CSV asset data</a></li>
</ul>
</div>
{{ block.super }}
{% endblock %}
```

-END-